Predicting Olympic Medal Counts

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Abstract

This study aims to develop a statistical framework for predicting the total number of medals won by each country at the 2016 Olympic Games, using historical performance and socio-economic data from 2000 to 2016. The analysis focuses on understanding how factors such as previous medal counts, economic strength, population, and political context influence Olympic success. To address the uneven distribution of medals across countries—where many win none and a few dominate—Principal Component Analysis (PCA) and a Zero-Inflated Negative Binomial (ZINB) model were applied.

Beyond the primary goal of predicting 2016 outcomes, the study also examined which variables contribute most strongly to medal success, how accurately gold medal results can be modelled, and whether historical trends can provide insight into future Olympic performance. Although long-term forecasting was limited by data availability, the overall findings show that combining past results with socio-economic analysis through statistical modelling offers a powerful and interpretable way to understand and anticipate global patterns of Olympic achievement.

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Chapter 1

Introduction

The Olympics is the premier sporting event in the world, drawing together thousands of athletes who spend years preparing for the chance to compete on its global stage. For many, an Olympic medal represents the highest possible achievement in sport, a moment that defines careers and inspires entire nations. But beneath this celebration of individual excellence lies a deeper structural reality: Olympic success is rarely just about talent. Medal tallies often mirror a country's wealth, infrastructure, population strength, and long-term investment in sport, turning the Games into an unexpected reflection of national development. Countries with extensive resources and sustained institutional support regularly dominate the podium, while others struggle simply to qualify. In this sense, Olympic performance becomes both a symbol of prestige and a revealing indicator of socio-economic capacity, highlighting how global inequalities shape outcomes even in the most celebrated sporting arena.

Predicting medal outcomes offers insights into these patterns but poses statistical challenges: data are *sparse* (many nations win none), *uneven* (a few dominate), and *multicollinear* (key indicators overlap). To address this, the study combines **Principal Component Analysis (PCA)** for dimensionality reduction with a **Zero-Inflated Negative Binomial (ZINB)** model suited to overdispersed, zero-heavy count data.

While earlier work has used Poisson, Negative Binomial, or machine-learning models, few combine socio-economic and historical factors within an interpretable, zero-inflated framework. This thesis fills that gap by developing a transparent statistical model to predict total Olympic medals for 2016 and extending it to gold medals and linear forecasts for 2028.

The study pursues several interconnected aims. It begins by examining how national socio-economic indicators—such as income, population size, and investment in sport—interact with historical performance to shape Olympic outcomes. Building on this foundation, the thesis develops a predictive framework that integrates Principal Component Analysis with a Zero-Inflated Negative Binomial model, allowing the analysis to handle multicollinearity, overdispersion, and the prevalence of nations that win no medals. The model's performance is then assessed in detail, focusing on overall fit, predictive accuracy, and the interpretability of key predictors when validated against the 2016 Olympic results. Finally, the investigation extends beyond total medals to include gold-medal predictions and a set of linear forecasts for 2028, focused on the United States, China, and India.

From these aims emerge a set of research questions guiding the analysis. The first explores which socio-economic and historical factors exert the strongest influence on medal outcomes. The second tests whether the Zero-Inflated Negative Binomial framework provides meaningful improvements over simpler count-data models. A third investigates how the results differ when moving from total-medal counts to gold-specific predictions. The final question considers the practical and methodological challenges that arise when the model is pushed beyond historical data to produce forward-looking forecasts.

Chapter 2

Data Description and Exploratory Analysis

2.1 Data Sources and Structure

This study draws on secondary data covering seven Summer Olympic cycles (2000–2016), provided by the University of Glasgow. The dataset combines country-level medal outcomes with a range of socio-economic, political, and demographic indicators, including variables such as Gross Domestic Product (GDP), population, and historical performance measures. The resulting dataset includes 108 countries and 44 features observed across seven Olympic years.

The dependent variables are the total number of medals and the number of gold medals won by each country in 2016. The independent variables include economic and demographic factors (GDP, population, political ideology), Olympic-related characteristics (host-nation status, number of athletes), and prior performance indicators (previous medal counts).

In the later stages of the project, the total medals won by India, the United States, and China in the 2020 and 2024 Olympic Games were obtained from the official Olympics website to extend the historical record used for forecasting.

To provide an overview of the data, Figure 2.1 presents a map visualisation of total medals from the 2016 Olympic Games. The bubble size represents the number of medals won by each country. It can be seen from the map that, apart from a few dominant nations such as the United States, China, and Russia, most countries have very few medals, with many recording none at all. This stark imbalance visually illustrates the near-zero inflation present in the dataset, where a large proportion of observations correspond to non-medal-winning nations.

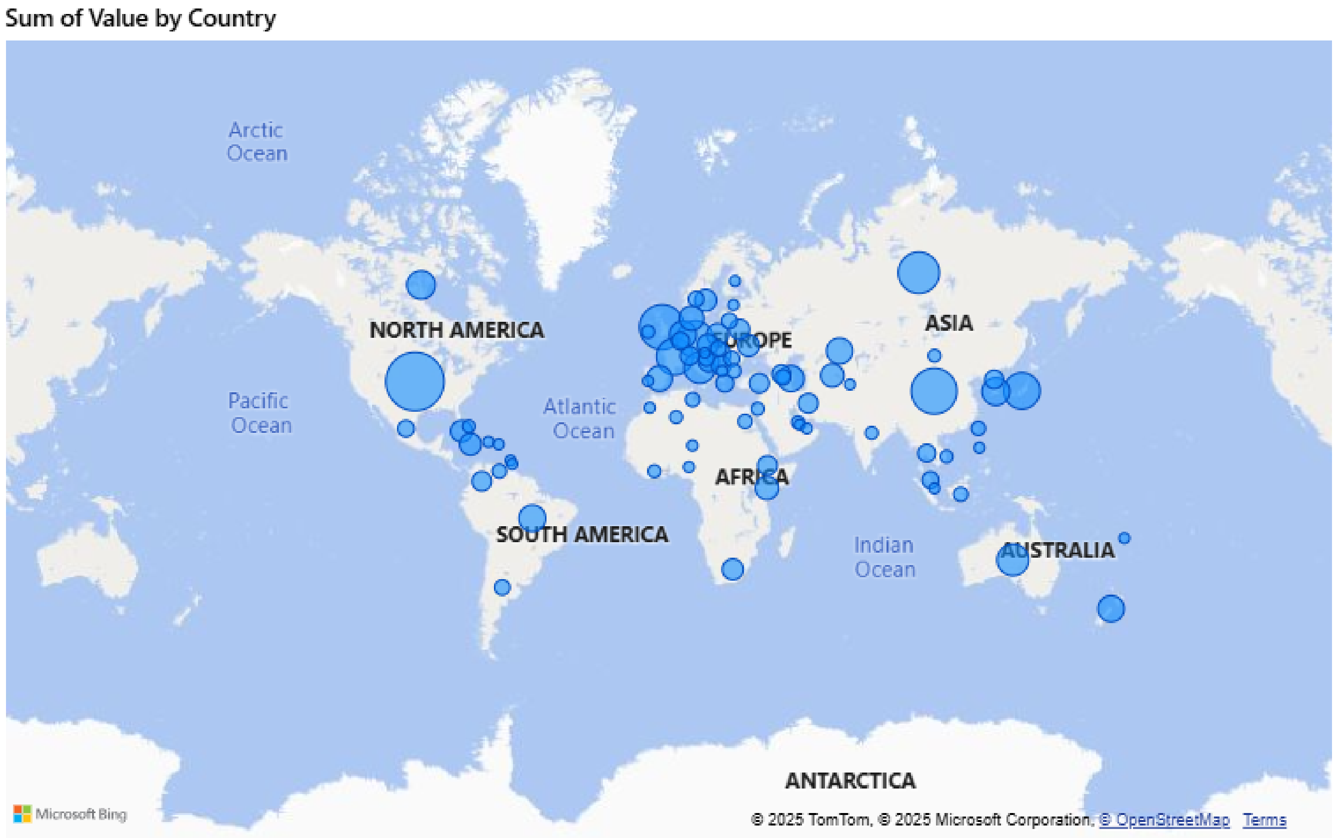

Figure 2.1: Total medals in 2016 per country

Data preprocessing proceeded in clearly defined steps. First, obviously non-informative fields (e.g., country name, country code, total medals achievable) were removed from the modelling matrix. Next, predictors were screened against the target variable tot16 (total medals in 2016): for continuous predictors I used Pearson correlations, and for categorical predictors one-way ANOVA, retaining only variables with $p \leq 0.01$. Insignificant variables such as athlete BMI, host-country status, and political ideology were excluded from the dataset which reduced the feature set from 44 to 25 relevant variables. From this reduced dataset, two missing values were identified and imputed using k-Nearest Neighbours (KNN) with $k = 5$, which preserves the multivariate relationships between countries while filling data gaps more effectively than simple mean substitution. The same screening pipeline was later repeated with gold medals in 2016 as the target. Applying identical thresholds and tests yielded 25 retained predictors for the gold-medal specification as well. Before formal modelling, exploratory analysis was carried out to understand the overall distribution of medals, the extent of zero inflation, and the interrelationships among predictors.

2.2 Exploratory Analysis I: Medal Distribution and Zero Inflation

To assess the suitability of different count-based models, the distribution of total medals in 2016 (`tot16`) was first examined. The mean medal count across 108 countries was **8.96**, while the variance was **287.29**, giving a variance-to-mean ratio of approximately **32.05**. This substantial difference confirms the presence of **overdispersion**, a condition where the variance greatly exceeds the mean. Since the Poisson model assumes equality between the mean and variance, it was immediately ruled out as inappropriate for this dataset.

In contrast, the **Negative Binomial (NB)** model allows the variance to exceed the mean by incorporating a dispersion parameter (α), which adjusts for unobserved heterogeneity. The variance of the NB model is given by:

$$\mathrm{Var}(Y_i) = \mu_i + \alpha\mu_i^2$$

where μ_i is the expected count for observation i, and $\alpha > 0$ represents the dispersion parameter. A larger α indicates greater overdispersion relative to the Poisson case.

However, before finalising the model choice, the dataset was further examined for zero inflation, a common feature in Olympic medal data where many countries record no medals at all. A pie chart of medal distribution (Figure 2.2) shows that **23.1%** of countries—that is, **25 out of 108 nations**—recorded zero medals in 2016. This indicates the possible presence of **zero inflation**, where the number of zero observations exceeds what the standard Negative Binomial model can account for.

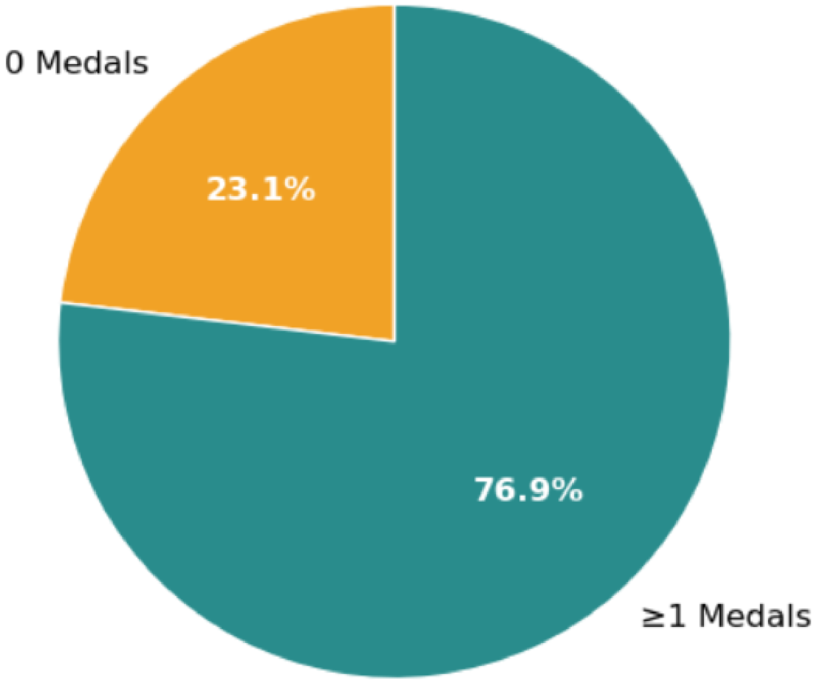

Figure 2.2: Proportion of countries with zero vs. non-zero medals.

In a **zero-inflated model**, the outcome is assumed to arise from two separate processes. The first process is a binary (logit) model that determines whether a country is a 'certain zero" (i.e., a nation that has no realistic chance of winning a medal). The second process models the count outcome (e.g., number of medals) for countries that participate competitively. This dual structure allows the model to better represent datasets where zeros occur for both structural reasons (countries lacking resources or representation) and stochastic reasons (chance outcomes among capable competitors).

To test whether the observed zeros exceeded what a standard Negative Binomial (NB) model could explain, the number of zeros that the NB distribution could theoretically accommodate was estimated using its dispersion parameter. For the NB model, the probability of observing a zero count is given by:

$$P(Y_i = 0) = \left(\frac{\alpha^{-1}}{\alpha^{-1} + \mu_i} \right)^{\alpha^{-1}}$$

where μ_i is the expected mean count and α is the dispersion parameter. The expected number of zeros in the sample is then obtained by multiplying this probability by the number of observations n:

$$E(0) = n \times P(Y_i = 0)$$

Substituting the estimated values of $\mu = 8.96$, $\alpha = 0.28$, and $n = 108$, the model predicts approximately $E(0) = 39.7$ zero observations. This value is slightly higher than the 25 actual zeros observed in the data. This comparison suggests that, although the data are clearly overdispersed, the degree of zero inflation is not extreme enough to make a zero-inflated specification strictly necessary.

However, as shown in the boxplot of total medals (Figure 2.3), most countries that do win medals record only single-digit counts, while a few dominate the higher ranges. This pattern indicates a form of **near zero inflation**, where the dataset contains a heavy concentration of low but non-zero values. Such distributions often arise in real-world count data when the underlying process produces a mixture of structural zeros (countries that almost never win medals) and a dense cluster of small counts (countries with limited but non-negligible success). In these situations, the **Zero-Inflated Negative Binomial (ZINB)** model remains advantageous for several reasons.

First, it explicitly models the excess probability mass near zero through its dual-process structure: one component governs the probability of belonging to a 'certain-zero" group, while the other models counts for those capable of winning medals. This separation allows the model to distinguish between nations that are *structurally zero* (those lacking resources or participation) and those that are *sampling zero* or low-count countries (those that could win medals but did not in a given year).

Second, even when the total number of exact zeros does not substantially exceed the expectation under the Negative Binomial model, the presence of a large number of small counts violates the smooth unimodal assumption of standard NB distributions. The ZINB model accommodates this by inflating the lower tail of the distribution, improving fit in datasets where near-zero observations are overrepresented. In practical terms, it captures the empirical asymmetry between countries that rarely win medals and those that dominate, reducing bias in parameter estimates and enhancing predictive accuracy for low-performing nations.

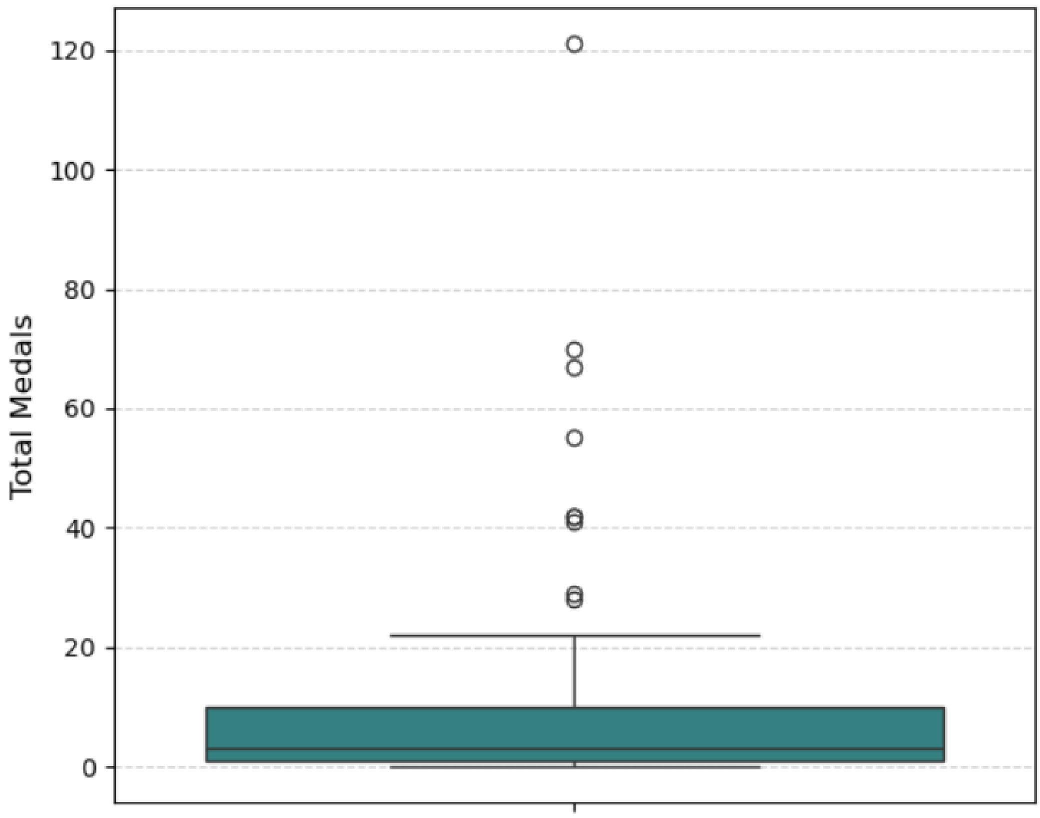

Figure 2.3: Boxplot of total medals across countries in 2016. The distribution is heavily right-skewed with many low-count observations and a few dominant outliers.

Together, these results demonstrate that Olympic medal data exhibit both **overdispersion** and **near-zero inflation**, validating the use of the ZINB framework for subsequent modelling. The next section examines correlations among predictors to address potential multicollinearity before proceeding to model development.

2.3 Exploratory Analysis II: Correlations and Multicollinearity

Following the initial screening, correlations among the remaining **25 variables** were examined to assess multicollinearity. Many socio-economic indicators are conceptually and statistically interrelated. In addition, the number of medals won in previous Olympics is strongly correlated with both GDP-related variables and current medal counts, indicating that national wealth and infrastructure are major drivers of Olympic success.

The correlation heatmap in Figure 2.4 shows strong correlation between these variables, confirming the presence of redundancy. High correlation between predictors can inflate standard errors and distort coefficient interpretation, making it essential to reduce dimensionality before modelling.

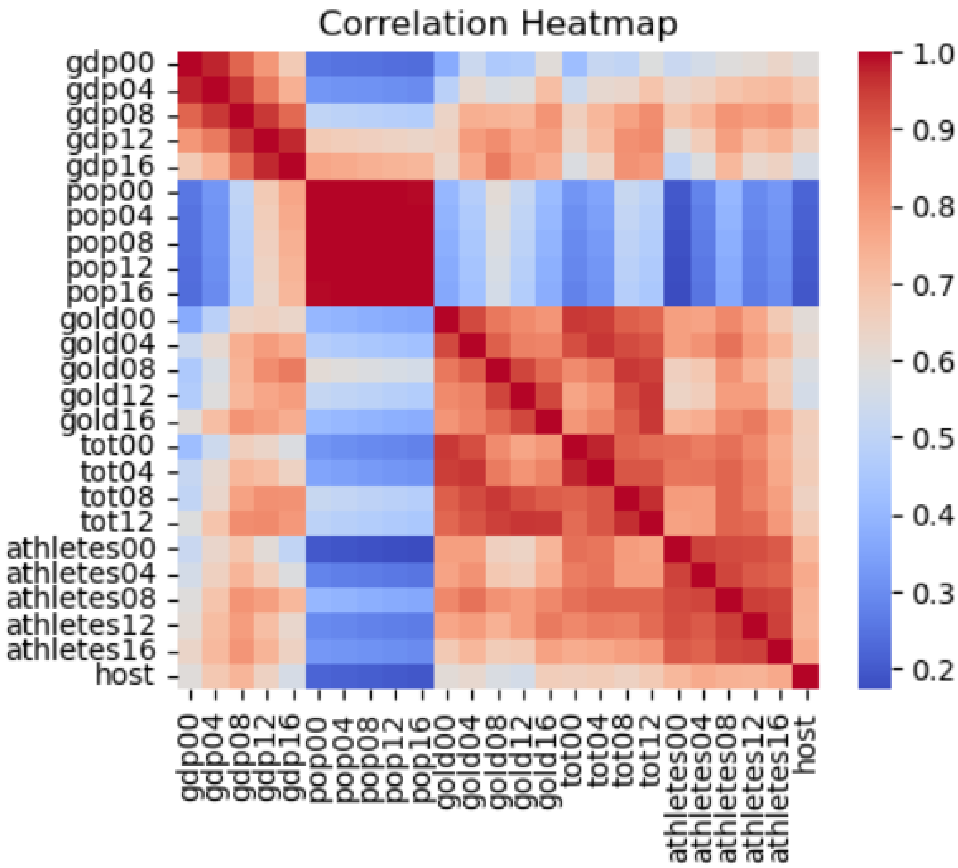

Figure 2.4: Correlation heatmap of predictors

To address this issue, the analysis employed **Principal Component Analysis (PCA)**, which transforms correlated variables into a smaller set of orthogonal components. This approach helps retain most of the explanatory power while removing redundant information, ensuring more stable and interpretable model estimation.

2.4 Principal Component Analysis

Principal Component Analysis (PCA) was applied to the 25 standardised predictors. The cumulative variance curve indicated a clear elbow and subsequent stabilisation at **six** principal components (Figure 2.5). On this basis, **PC1–PC6** were selected for the modelling stage. To check robustness, the model was re-estimated using alternative component counts (**2, 3, 4, 5, and 7** PCs). Across these specifications, the six-component solution consistently delivered the best balance between fit and parsimony—improving predictive performance relative to fewer components, while avoiding the diminishing returns and added complexity observed beyond six.

Qualitatively, the retained components capture broad socio-economic axes while effectively removing multicollinearity among the original variables. This dimensionality reduction provides a stable and interpretable input set for the subsequent count models.

The first six principal components capture the main patterns underlying the socio-economic and historical Olympic data. PC1 reflects a broad "national capacity" dimension, with consistently positive loadings across GDP, population, athlete numbers, and past medal counts. PC2 contrasts population size with medal efficiency, separating highly populated but low-performing nations from smaller countries with stronger Olympic records. PC3 emphasises early-2000s economic variation relative to medal outcomes, while PC4 highlights differences in athlete delegation size and mid-period performance. PC5 distinguishes countries with stronger gold-medal concentrations from those with larger overall medal totals. PC6 is dominated by the host variable, indicating that hosting the Games contributes a distinct source of variation independent of broader socio-economic factors.

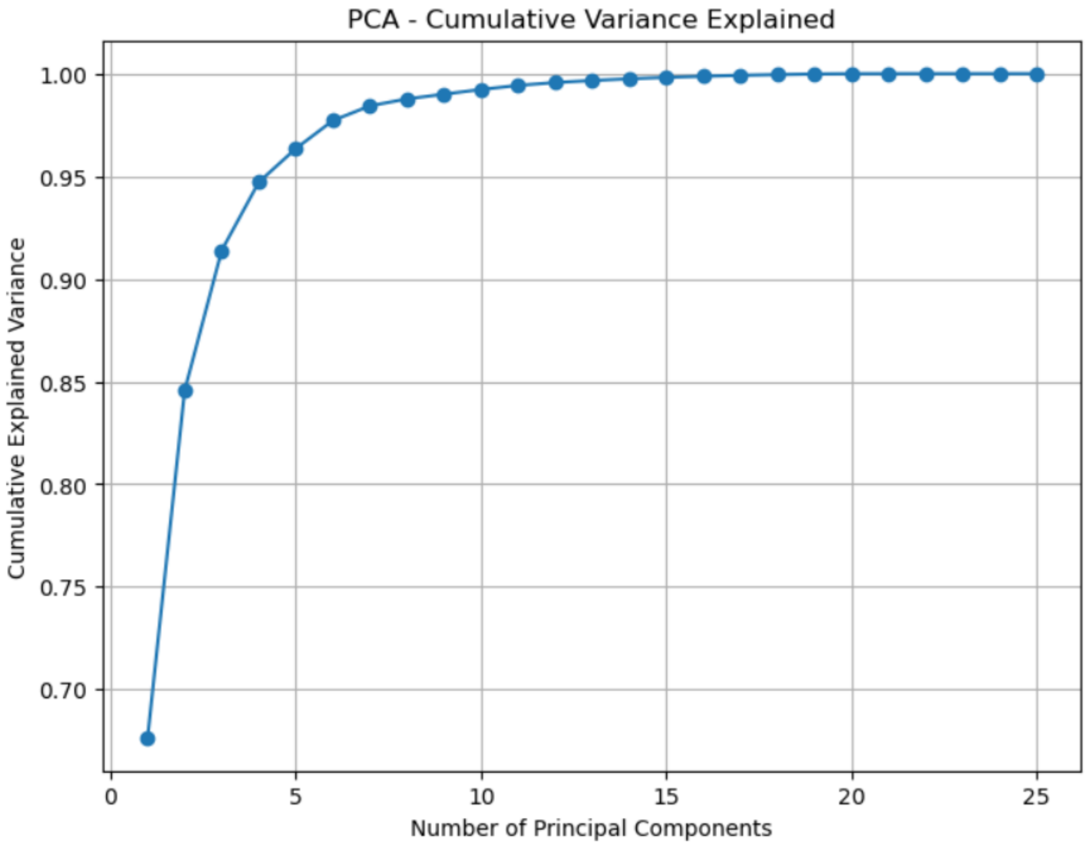

Figure 2.5: Cumulative variance explained by principal components. The curve stabilises at PC6, indicating the optimal number of components for modelling.

Later in the project, all these steps were repeated using **gold medals in 2016** as the target variable. The results were consistent with the total-medal analysis: **zero inflation** and **overdispersion** were again observed, **25 variables** were retained after significance screening, and **six principal components** were selected based on the same variance-stabilisation criterion. The only major difference was that, in this case, the standard Negative Binomial model failed to account for the higher number of zero counts, confirming the presence of **true zero inflation** and reinforcing the need for the Zero-Inflated Negative Binomial (ZINB) specification.

Chapter 3

Methodology

3.1 Model Specification

Two count-based models were constructed to predict the total number of medals won by each country in the 2016 Olympic Games: a **Negative Binomial (NB)** model and a **Zero-Inflated Negative Binomial (ZINB)** model. Both models utilised the six principal components (PC1–PC6) derived from the 25 standardised predictors as explanatory variables. The dataset was divided into training and testing subsets using an 80–20 split to evaluate predictive performance. The United States was excluded from model fitting due to its extreme outlier status, as evident from the boxplot of total medals. Its inclusion caused both models to significantly overshoot predictions for other countries, reducing overall accuracy.

3.1.1 Negative Binomial Model

The Negative Binomial model is appropriate for overdispersed count data where the variance exceeds the mean. The model assumes that the conditional mean of the response variable Y_i (the medal count for country i) depends on a linear combination of predictors through a log link.

The probability distribution of the Negative Binomial (NB) model is given by:

$$P(Y_i = y_i) = \frac{\Gamma(y_i + \alpha^{-1})}{\Gamma(\alpha^{-1})\, y_i!} \left(\frac{\alpha^{-1}}{\alpha^{-1} + \mu_i}\right)^{\alpha^{-1}} \left(\frac{\mu_i}{\alpha^{-1} + \mu_i}\right)^{y_i}, \quad y_i = 0, 1, 2, \ldots$$

where μ_i is the expected medal count for country i, and $\alpha > 0$ is the dispersion parameter controlling the degree of overdispersion relative to the Poisson model.

3.1.2 Zero-Inflated Negative Binomial Model

To account for excess zeros in the medal data, a Zero-Inflated Negative Binomial (ZINB) model was also fitted. The ZINB model assumes that zeros can arise from two distinct processes: (i) a *logit model* that generates "certain zeros" for countries with negligible probability of winning a medal, and (ii) a *count model* (Negative Binomial) that describes medal counts for competitive countries.

The probability distribution of the ZINB model is given by:

$$P(Y_i = 0) = \pi_i + (1 - \pi_i) \left(\frac{\alpha^{-1}}{\alpha^{-1} + \mu_i}\right)^{\alpha^{-1}}$$

$$P(Y_i = y_i) = (1 - \pi_i) \frac{\Gamma(y_i + \alpha^{-1})}{\Gamma(\alpha^{-1})\, y_i!} \left(\frac{\alpha^{-1}}{\alpha^{-1} + \mu_i}\right)^{\alpha^{-1}} \left(\frac{\mu_i}{\alpha^{-1} + \mu_i}\right)^{y_i}, \quad y_i > 0$$

Here, π_i represents the probability that observation i belongs to the zero-generating process, estimated through a logistic regression that determines whether a country is classified as a structural zero rather than a sampling zero. Formally, the zero-inflation probability is given by:

$$\text{logit}(\pi_i) = \mathbf{z_i}'\gamma$$

Here, $\mathbf{z_i}$ denotes the predictors used for the zero-inflation component (set to the same six PCs for simplicity), and γ is the associated parameter vector.

Model performance was evaluated using prediction accuracy. The ZINB model demonstrated a better fit than the standard NB model, confirming that the zero-inflation component improved explanatory power and prediction accuracy.

3.2 Gold Medal Modelling

After confirming the superiority of the ZINB specification, the same modelling pipeline was applied using **gold medals in 2016** as the dependent variable. The same six principal components were used as predictors. Results showed similar behaviour to the total-medal model, with overdispersion and excess zeros again present.

3.3 Forecasting Medals for 2028

The final stage of analysis aimed to forecast total medals for the **2028 Olympic Games** for three countries—**the United States, China, and India**. Historical medal counts for the years 2000, 2004, 2008, 2012 and 2016 were sourced from the university provided dataset and 2020 and 2024 were obtained from the official Olympics website.

Given the limited number of observations ($n = 7$), more complex time-series models such as ARIMA or polynomial regression were tested but found to overfit the data. Consequently, a simple **linear regression model** was chosen for its stability and interpretability. The model for each country is defined as:

$$Y_t = \beta_0 + \beta_1 t + \varepsilon_t$$

where Y_t denotes the total medal count in year t, and ε_t is a random error term. The fitted linear trends were then extrapolated to predict expected medal counts for the 2028 Olympics.

This approach balances parsimony and interpretability, providing reasonable forecasts without imposing overly complex dynamics on limited historical data.

Before model fitting, the United States was excluded from the training dataset. Its exceptionally high medal counts made it an extreme outlier, causing the model to overshoot predictions for mid-performing nations and inflate error metrics. Excluding the USA stabilised the estimation process and improved generalisation for the majority of countries.

Chapter 4

Results and Conclusion

4.1 Model Comparison and Fit

Two models were initially fitted to the total-medal dataset using the six principal components derived from PCA: a **Negative Binomial (NB)** model and a **Zero-Inflated Negative Binomial (ZINB)** model. Model performance was evaluated on the test set using the Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE). Table 4.1 summarises the results.

Table 4.1: Model comparison on the test set

Model	**MAE**	**MSE**	**RMSE**	**Convergence**
Negative Binomial	64.531	81222.122	284.995	Non-robust
Zero-Inflated Negative Binomial	**3.329**	**21.125**	**4.596**	No

The ZINB model substantially outperformed the standard Negative Binomial model across all error metrics, achieving an MAE of 3.33 and an RMSE of 4.60. In practical terms, this means that on average, medal counts in the test set were predicted within ± 4.6 of the actual values. Although the ZINB estimation reported a non-convergence warning, repeated runs with alternative starting values and also with gold medals as the target variable yielded stable results. The minimal change in MAE and RMSE across iterations indicates that the optimisation is oscillating around the global minimum rather than diverging, suggesting that the model is in the vicinity of the ideal parameter estimates.

4.2 Prediction Accuracy and Visualisation

Figure 4.1 plots the predicted versus actual total medal counts for the test set. The model performs robustly across the full range of counts.

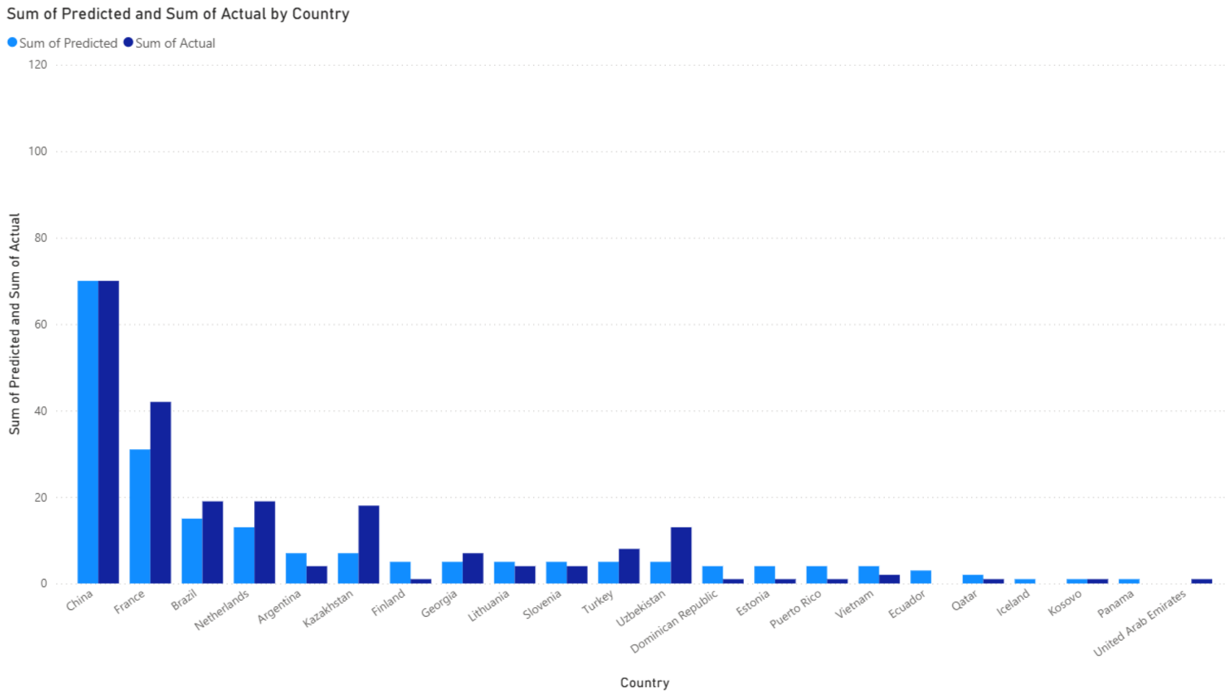

Figure 4.1: Predicted versus actual medal counts (test set) for total medals.

4.3 Gold-Medal Model Evaluation

The same PCA–ZINB pipeline was repeated with **gold medals in 2016** as the target variable. The model achieved an MAE of 1.056, MSE of 1.916, and RMSE of 1.384, confirming a strong fit even under greater zero inflation. These low error values demonstrate that the model generalises well across different count scales and that the structural assumptions of the ZINB framework are appropriate for Olympic medal data.

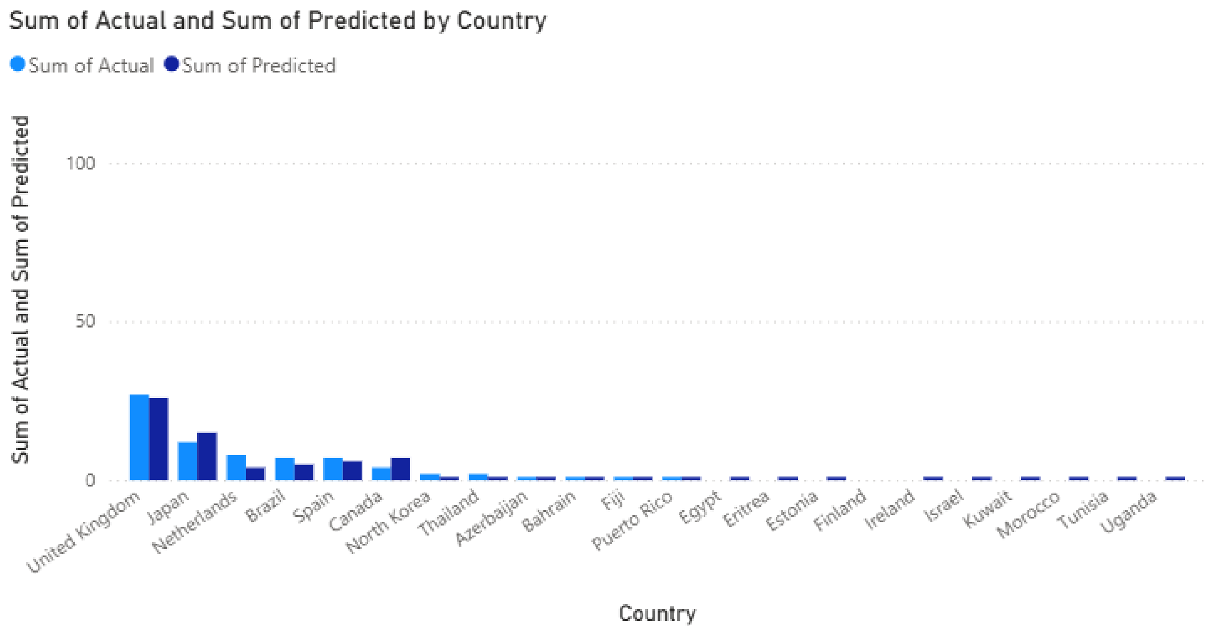

Figure 4.2: Predicted versus actual gold medals (test set).

4.4 Forecasting Results for 2028

Using the fitted framework, total medal counts for the United States, China, and India were extended through a simple linear-trend regression model based on historical outcomes from 2000 to 2024. Since only seven data points were available for each nation, more complex models such as polynomial regression or ARIMA tended to overfit and were therefore rejected. The linear model offered the most stable extrapolation.

The resulting forecasts for the 2028 Los Angeles Olympics are summarised below:

Table 4.2: Forecasted total medals for 2028 (Linear Regression)

Country	**Predicted Total Medals (2028)**
United States	128.71
China	97.57
India	7.43

Figure 4.3 visualises the linear trends alongside historical data. The United States and China show stable or gradually increasing trends, while India exhibits a modest upward trajectory, consistent with its growing sporting investment. These forecasts align with expectations based on national resource patterns and previous Olympic trajectories.

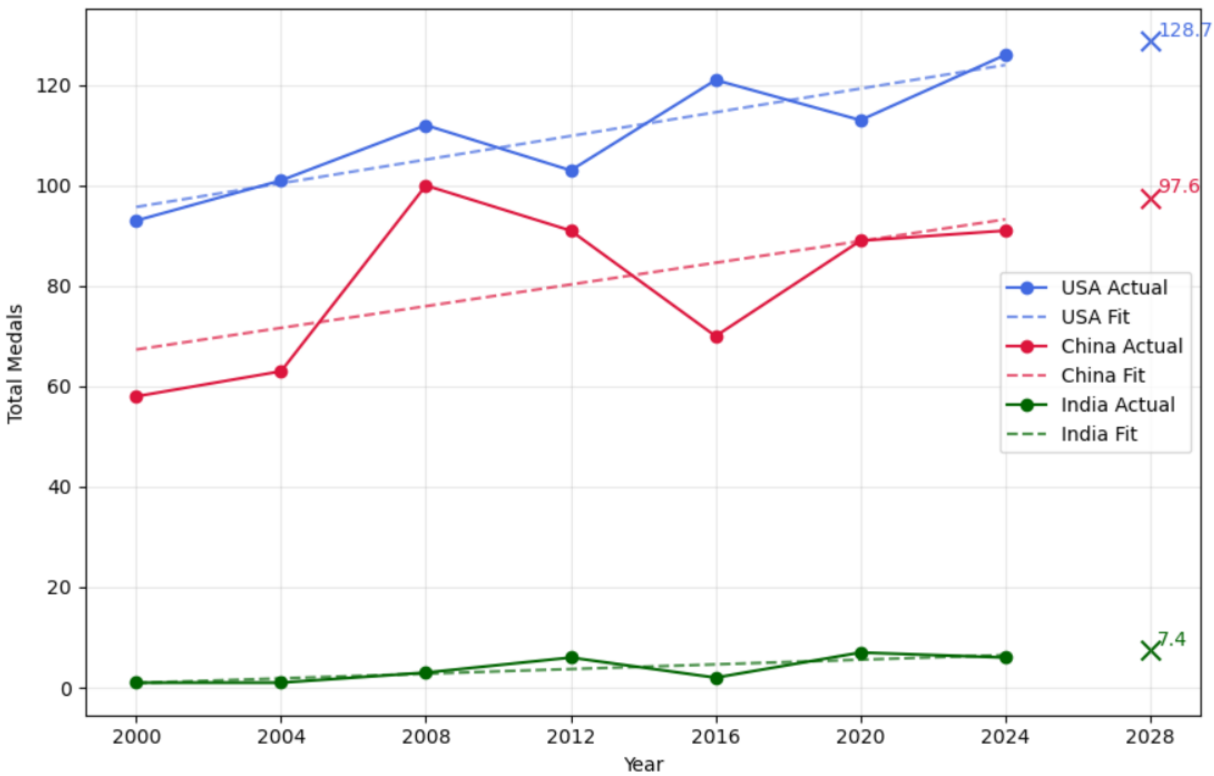

Figure 4.3: Linear regression forecasts for USA, China, and India (2000–2028).

4.5 Limitations and Future Work

Attempts were made to extend the ZINB model to forecast medal counts for the 2028 Olympic Games. However, data collection proved to be a significant challenge. Many socio-economic indicators were incomplete, inconsistent across sources, or available only as estimates, which introduced uncertainty into the modelling process. As a result, the extended ZINB model did not perform reliably when applied to future projections. Another limitation of the present study is the relatively small number of explanatory variables, which restricts the model's ability to capture the full complexity of factors influencing Olympic success. Future work could focus on expanding and refining the dataset through direct collaboration with official Olympic data repositories, ensuring accurate and consistent socio-economic indicators across time. With a more comprehensive and verified dataset, the existing framework could be extended to provide robust and interpretable medal forecasts for upcoming Olympic Games.

4.6 Discussion and Conclusion

The findings of this study highlight the strong relationship between a nation's economic development and its Olympic success. Among all predictors, GDP and past medal performance emerged as the most influential predictors in the Pearson and ANOVA significance tests, reflecting a clear sense of inertia—countries that have historically performed well continue to do so, while resource-limited nations struggle to catch up. This persistence underscores how wealth and institutional investment create lasting advantages in global sport. The forecasting analysis further supports this pattern: the United States remains projected to lead overall medal counts, with China continuing to narrow the gap, and India showing gradual improvement from a minimal to a moderate-performing nation. These results suggest that although emerging economies can improve their standings through sustained investment, large structural shifts in Olympic dominance are likely to remain rare.

This study also demonstrates that Zero-Inflated models can perform effectively even under near-zero inflation, capturing subtle asymmetries in low-count data while maintaining interpretability. The framework developed here can be generalised to other overdispersed count datasets beyond sports analytics, offering a flexible approach to modelling outcomes with structural and sampling zeros.

In conclusion, the integration of PCA and ZINB provides both predictive strength and interpretive clarity, showing how rigorous statistical modelling can illuminate broader patterns of inequality and persistence within global competition.

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